Smart Brewery - Project Proposal
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Problem Identification

When homebrewers do not use a smart electric brewery, they face various challenges that could impact the quality and consistency of their brewing process. The absence of automation means relying on manual tasks, which can result in errors and inefficiencies. Additionally, the lack of data logging features makes it challenging to troubleshoot problems and fine-tune recipes. Furthermore, without remote monitoring capabilities, brewers are limited in their ability to adjust and oversee the brewing process, reducing their option to re-create their own favorite beer recipes.

Solution Overview

The Smart Brewery Project aims to modernize home brewing through IoT and real-time analytics. By utilizing smart sensors for brewing temperature, fermentation, and ingredient control, it enhances brew quality and flavor consistency. Home brewers can automatically manage their brewing operations, receive alerts, access insights about their brewing recipe history files, and receive AI recipe suggestions. This project represents a fusion of traditional craft with data-driven efficiency, promising a transformative impact on the home brewing industry by improving the brewing process and overall experience.

Existing Alternatives

Manual cooking: Using basic tools like a temperature sensor and gas heat to regulate temperature. Brewers monitor and adjust temperature manually, requiring constant attention for consistency and great effort to maintain the wanted temperature.

Cooler Brewing Method: This method involves using a cooler, such as a large insulated container, to maintain specific temperatures during the mashing process. Brewers can heat water to a precise temperature and then transfer it to the cooler along with the grains, allowing them to mash at a consistent temperature over time.

Immersion Circulator: An immersion circulator, commonly used in sous vide cooking, can also be employed for precise temperature control during brewing. By circulating water at a precise temperature around the brew vessel, brewers can achieve consistent temperatures throughout the mashing and fermentation processes.

Electric Heating Elements: Instead of gas, brewers can utilize electric heating elements coupled with temperature sensor control to regulate the brewing temperature. This approach provides more precise temperature control compared to traditional gas heating methods.

User Demographics

The expected user base of the Smart Brewery application can be diverse, catering to individuals with varying levels of brewing experience and interests in both traditional craft brewing and modern technological advancements. The target audience includes:

- Individuals who are new to homebrewing and are looking for an accessible and user-friendly platform to start their brewing journey.
- Home brewers with some experience who seek to enhance their brewing skills and experiment with different recipes using smart technology.
- Those passionate about craft beer who appreciate the fusion of traditional brewing techniques with modern technology for improved quality and consistency.
- Enthusiasts who are interested in incorporating IoT and real-time analytics into their brewing process, leveraging smart sensors and automation for precise control.
- Individuals who enjoy sharing their brewing experiences, recipes, and tips within a community, fostering a collaborative environment for knowledge exchange.
- Those who are open to exploring AI-generated recipe suggestions, adding a layer of innovation to their brewing experience and encouraging experimentation with new flavors.

The Smart Brewery application aims to appeal to a broad range of users by offering a seamless blend of traditional brewing practices and advanced technology, making home brewing more accessible, enjoyable, and efficient for a diverse audience.

Dependencies

Embedded:

Development languages: Python, C++

FrameWorks: RaspberryPi Os, Arduino IDE

Internet Connectivity: wifi

Api Communication: HTTP microservices

Back-End:

Development language: Java

FrameWork: Maven, Spring Boot

Data Base: PostgreSQL

Api Communication: HTTP microservices

Front-End:

Development language: JS

FrameWork: React

Application: Web

Api Communication: HTTP

Data Source:

1. https://www.kaggle.com/datasets/ankurnapa/brewery-operations-and-market-analysis-dataset/data
2. https://www.kaggle.com/datasets/basaltier/homebrew-beer-data

Features and User Flows:

General:

- **Sync Feature:** Enable effortless synchronization between the app and the heat unit's temperature sensor, as well as the fermentation sensor, using Bluetooth or WiFi connectivity. This ensures real-time data exchange, allowing the app to automatically update the cloud based server with the latest temperature and fermentation information gathered from the connected devices. Users can thus easily monitor and control brewing conditions remotely, fostering a user-friendly and efficient brewing experience.
- **User Accounts Management feature:** Users need to have individual accounts to access and save personalized brewing recipes.
  Functions:
    1. Create new accounts
    2. Update account data
    3. Delete accounts
- **Notification Feature:** Receive instant updates on your brewing process with the notification feature, alerting you to crucial milestones, temperature adjustments, and fermentation progress, ensuring you stay informed and in control of every aspect of your beer-making journey.
- **Performance:** The app should deliver optimal performance, providing users with swift access to real-time brewing data, intuitive controls, and seamless synchronization with brewing devices, ensuring a smooth and efficient homebrewing experience.
- **Security:** The app should prioritize security by implementing robust encryption protocols to safeguard sensitive brewing data, ensuring a secure and confidential environment for users to manage their home brewing processes.
- **Reliability:** The app should be reliable and stable, experiencing minimal crashes or bugs. Additionally, it should have the capability to recover from errors and provide helpful messages to users.

Share & Download Recipes Online:

- **User Profiles:** Enable user profiles for individuals to contribute and share their recipes to their followers who can download those recipes to their “favorites”.
- **User Contribution:** Allow users to contribute, rate, and review recipes, fostering a collaborative community.

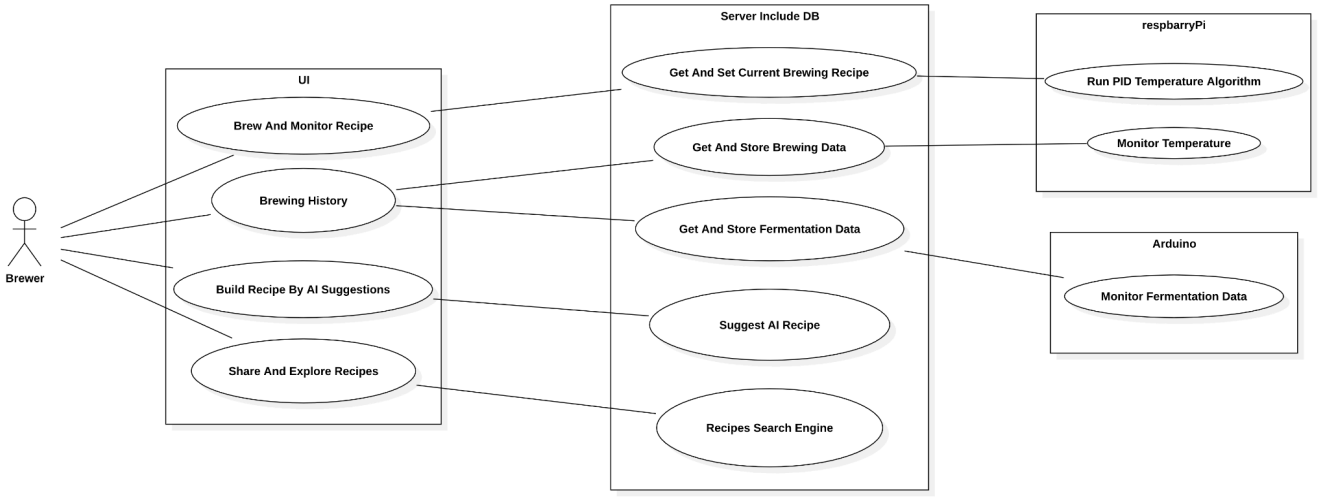
Brew a Recipe:

- **Recipe Input:** A user-friendly interface for users to input and customize brewing recipes.
  Functions:
  1. Create new recipe
  2. Update an existing recipe
  3. Delete recipe
- **Automated System:** Integration of an iot automated brewing system monitoring temperature sensor and controlling heat unit, capable of executing recipes with a single click.
- **Control Interface:** An intuitive control interface to provide users with easy control over the brewing process.
  Functions:
  1. Show the recipe stage
  2. Show the current temperature
  3. Show the remaining time of the current stage/entire process
  4. Show fermentation data

Brewing History:

- **Data Logging:** Implement a feature for the system to log relevant data from each brewing session into a database.
- **Search Functionality:** A search functionality for users to efficiently locate and review specific brewing sessions, sorted by name, oldest/newest, rate, etc.
- **Analytics Module:** Integration of an analytics module to provide insights and trends from brewing history.
- **Brewing Taste Result:** Establish a mechanism for users to input and update their brewing flavors result.

Get AI Beer Recipes:

- **AI Integration:** Implement an artificial intelligence system capable of crafting diverse and customized beer recipes by customized parameters.
- **Data inputs:** Using the user's brewing history and taste rates results data to run the AI engine for beer recipes recommendations.

Use-Case Diagram